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# Chemical Thermodynamics

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Spontaneous → Those process occurs on its own without the help of external agency. It is also a natural process.  
for eg:

- a) roasting of iron
- b) flow of water from higher level to lower level.
- c) flow of heat from higher temp<sup>r</sup> to lower temp<sup>r</sup> of a body.
- d) mixing of gases etc.

## Characteristics of Spontaneous Process

- a) It is unidirectional in nature (ie irreversible)
- b) Some spontaneous process needs initiation due to high value of activation energy.  
eg burning of fuel.
- c) Spontaneous process leads system towards increase in randomness.
- d) It may be slow or fast.

Non-spontaneous process → Those process occurs only with the help of external agency is non-spontaneous. It is also an artificial process.

eg:

- a) separation of  $O_2$  from atmospheric air.
- b) flow of water from lower level to higher level.
- c) flow of heat from colder body to hotter body.



## # Enthalpy change & Spontaneity of process.

Exothermic reaction proceeds with release of heat and all exothermic rxn such as burning of fuel etc are spontaneous in nature i.e.

If  $\Delta H = -ve$ , process is spontaneous.

For endothermic rxn, some process such as melting of ice, condensation of water vapour etc. are spontaneous whereas other process such as decomposition of calcium carbonate, formation of ammonia etc are non-spontaneous. i.e.

If  $\Delta H = +ve$ , process is non-spontaneous or may be spontaneous.

So, enthalpy change only cannot determine the spontaneity of process or enthalpy change is not the sole criteria for spontaneity of reaction.

## # Entropy (S):

To explain spontaneous process and to state second law of thermodynamics, Entropy is introduced. Entropy is defined as the measure of degree of randomness or disorderliness of the system. More random or disorder in the system, more will be the entropy. For eg: when a drop of ink is introduced into a glass of drinking water, it spreads spontaneously and makes the system disorder i.e. water can't be drinkable.

Entropy has important physical significance. The value of entropy of solid is less. Since the molecules of solid are ordered properly (i.e. less random), in case of liquid the molecules are





arranged more randomly than solid so that entropy of liquid is more than solid whereas in case of gas, molecules are randomly ~~order~~ arranged. Hence it has very high ~~no~~ value of entropy.

$$S_{\text{solid}} < S_{\text{liquid}} < < < S_{\text{gas}}$$

Entropy is a state function and also an extensive property. Mathematically,

where,  $S = \text{Entropy}$

$$S = k \ln W$$

$k = \text{Boltzmann's constant}$

$W = \text{probability factor}$

Quantitatively,

$$\Delta S = \frac{q_{\text{rev}}}{T} \text{ or } \frac{\Delta H}{T}$$

where,

$q_{\text{rev}} = \text{heat change reversibly}$

$T = \text{working temp}^{\circ} \text{ in Kelvin}$

## # Entropy change and spontaneity of process

A spontaneous process leads the system towards increase in disorderness or randomness of a system i.e. increase in entropy. If there is increase in entropy the process is spontaneous for exothermic as well as endothermic otherwise not.  
i.e.

If  $\Delta S = +ve$ , process is spontaneous.

If  $\Delta S = -ve$ , process is non-spontaneous

If  $\Delta S = 0$ , process is in equilibrium.



## # Entropy of Surrounding (for spontaneous process)

For exothermic spontaneous process, Entropy of surrounding is given by

$$\Delta S_{\text{surrounding}} = \frac{-\Delta H_{\text{system}}}{T} \quad (\text{on the basis of system})$$

## Entropy of universe and spontaneity process

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

For exothermic spontaneous process

$$\Delta S_{\text{univ}} = \frac{-\Delta H_{\text{sys}}}{T_{\text{sys}}} + \left( \frac{\Delta H}{T} \right)_{\text{surr}}$$

Here,  $T_{\text{sys}} > T_{\text{surr}}$ , so  $\left( \frac{-\Delta H}{T} \right)_{\text{sys}} < \left( \frac{\Delta H}{T} \right)_{\text{surr}}$

So,  $\Delta S_{\text{surr}} = +ve$

For endothermic

$$\Delta S_{\text{univ}} = \left( \frac{+\Delta H}{T} \right)_{\text{sys}} + \left( \frac{\Delta H}{T} \right)_{\text{surr}}$$

Here,  $T_{\text{surr}} > T_{\text{sys}}$ , so  $\left( \frac{\Delta H}{T} \right)_{\text{sys}} > \left( \frac{\Delta H}{T} \right)_{\text{surr}}$

and  $\Delta S_{\text{univ}} = +ve$



Hence, If  $\Delta S_{\text{univ}} = +ve$ , process is spontaneous

If  $\Delta S_{\text{univ}} = -ve$ , process is non-spontaneous

If  $\Delta S_{\text{univ}} = 0$ , process is in equilibrium.

## # Entropy change in phase transformation.

- No heat change, No change in entropy

$$\Delta S = \frac{\Delta H}{T}$$

### a) Entropy of fusion ( $\Delta S_f$ )

The change in heat or entropy when 1 mole of solid substance changed in liquid state at its melting point is called entropy of fusion.

Mathematically,

$$\Delta S_f = S_{\text{liquid}} - S_{\text{solid}} \\ = +ve.$$

OR

$$\Delta S_f = \frac{\Delta H_f}{T_m} \leftarrow \text{melting temp}^{\circ}$$

### b) Entropy of vapourization ( $\Delta S_v$ )

The change in entropy when 1 mole of liquid substance changed into vapour state at its boiling point,

Mathematically



$$\Delta S_v = S_{\text{gas}} - S_{\text{liquid}}$$

$$= +ve.$$

OR

$$\Delta S_v = \frac{\Delta H_v}{T_b}$$

$T_b \leftarrow$  boiling temp<sup>r</sup>

c) Entropy of sublimation ( $\Delta S_s$ )

$$\Delta S_s = S_{\text{gas}} - S_{\text{solid}} = +ve$$

OR

$$\Delta S_s = \frac{\Delta H_s}{T_s}$$

$T_s \leftarrow$  sublimation temp<sup>r</sup>

d) Entropy of transition ( $\Delta S_t$ )

$$\Delta S_t = \frac{\Delta H_t}{T_t}$$

$T_t \leftarrow$  transition temp<sup>r</sup>.

e) Entropy of a reaction ( $\Delta S_{\text{rxn}}$ )

$$\sum \Delta S_{\text{all products}} - \sum \Delta S_{\text{all reactants}}$$

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## Second law of thermodynamics.

To overcome limitation of 1<sup>st</sup> law, define spontaneous reaction process and to state second law, entropy is introduced.

Second law states that, "for a spontaneous process the total entropy of the universe is always increasing i.e.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surrou.}} > 0$$

Other definition / statements.

- i) Clausius : It states that, "it is impossible for a cyclic process to transfer heat from body at lower temp<sup>r</sup> to the body at higher temp<sup>r</sup> unless some external work is done."
- ii) Lord Kelvin : It states that, "it is impossible to convert all the heat energy taken from the hot reservoir completely into workdone unless some part of heat is lost to the cold reservoir for a cycle process."

### Thermodynamic efficiency ( $\eta$ )

It is defined as the fraction of workdone from total heat gained to the total heat gained by the system.

Thus,

$$\eta = \frac{\text{Workdone (W)}}{\text{heat gained (q)}}$$

OR,

$$\eta = \frac{q_2 - q_1}{q_2}, \quad \eta = \frac{T_2 - T_1}{T_2}$$

Where,  $q_2$  = heat gained from hot reservoir

$q_1$  = heat lost " cold reservoir



$T_2$  = temp<sup>n</sup> of hot  
 $T_1$  = temp<sup>n</sup> of cold.

### Gibb's Free energy (G):

It is the more imp thermodynamic property which relates enthalpy, entropy and temp<sup>n</sup>. Gibb's free energy is defined as the useful form of energy associated with the system which can be completely converted into work done.

Thus,

$$G = H - TS$$

where,

$G$  = Gibb's free energy

$H$  = enthalpy of system

$T$  = temperature of

$S$  = entropy of system

Since Gibb's Free energy is state function and extensive property, change in free energy ( $\Delta G$ ) can be determined.

Let  $G_1, H_1, T_1, S_1$  and  $G_2, H_2, T_2, S_2$  be initial and final Gibb's free energy, enthalpy, temp<sup>n</sup> and entropy respectively i.e.

$$G_1 = H_1 - T_1 S_1 \quad \text{--- (1)}$$

$$G_2 = H_2 - T_2 S_2 \quad \text{--- (2)}$$

Subtracting eq<sup>n</sup> (1) from (2)

$$G_2 - G_1 = H_2 - H_1 - T_2 S_2 + T_1 S_1$$



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$$\Rightarrow \Delta G = \Delta H - (T_2 S_2 - T_1 S_1)$$

$$\Rightarrow \Delta G = \Delta H - \Delta(T \cdot S)$$

$$\boxed{\Delta G = \Delta H - T \cdot \Delta S - S \Delta T} \quad \text{--- (3)}$$

At constant temp<sup>r</sup>,  $\Delta T = 0$

$$\boxed{\Delta G = \Delta H - T \cdot \Delta S} \quad \text{--- (4)}$$

This is the mathematical relation of Gibbs free energy also known as 'Gibbs Helmholtz equation'.

(entropy)  
Imp Gibbs free change and spontaneity of process

For spontaneous process, From 2<sup>nd</sup> law of thermodynamics,

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

$$\Rightarrow \Delta S_{\text{univ}} = \Delta S_{\text{sys}} - \frac{\Delta H_{\text{sys}}}{T_{\text{sys}}} \quad \left( \because \Delta S_{\text{surr}} = -\left(\frac{\Delta H}{T}\right)_{\text{sys}} \right)$$

$\Rightarrow \Delta S_{\text{univ}}$  The subscript in R.H.S is only related to the system and hence can be omitted

$$\Delta S_{\text{univ}} = \Delta S - \frac{\Delta H}{T}$$

$$\Rightarrow \Delta S_{\text{univ}} = \frac{T \Delta S - \Delta H}{T}$$

$$\Rightarrow T \cdot \Delta S_{\text{univ}} = T \Delta S - \Delta H$$

$$\Rightarrow \Delta H - T \cdot \Delta S = -T \cdot \Delta S_{\text{univ}}$$



From Gibb's free energy

$$\Delta G = \Delta H - T\Delta S$$

$$\therefore \Delta G = -T\Delta S_{\text{univ}}$$

$$\boxed{-\Delta G = T\Delta S_{\text{univ}}}$$

Hence decrease in free energy relates spontaneous process. Hence,

- a) If  $\Delta G = -ve$ , process is spontaneous ( $T\Delta S_{\text{univ}} = +ve$ )
- b) If  $\Delta G = +ve$ , process is non-spontaneous ( $T\Delta S_{\text{univ}} = -ve$ )
- c) If  $\Delta G = 0$ , process is equilibrium.

To predict spontaneous process, free energy change is suitable since it relates all parameters of system only.

### Imp Gibb's free energy and useful workdone

From 1<sup>st</sup> law of thermodynamic

$$q = \Delta E + W \quad \text{--- (1)}$$

where,

$q = \text{heat}$

$E = \text{internal energy}$

$\Delta E = \text{change in enthalpy}$

$W = \text{total workdone}$

We know

$$W = W_{\text{mechanical}} + W_{\text{useful}}$$

$$W = P\Delta V + W_{\text{useful}} \quad \text{--- (2)}$$



From eqn (2) eqn (1) gives,

$$q = \Delta E + P \cdot \Delta V + W_{\text{useful}} \quad (3)$$

From enthalpy change,

$$\Delta H = \Delta E + P \cdot \Delta V \quad (4) \quad \text{At constant pressure}$$

From eqn (3) and (4)

$$q = \Delta H + W_{\text{useful}} \quad (5)$$

From Entropy change,

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$$\therefore q = T \cdot \Delta S \quad (6)$$

From eqn (5) and (6)

$$T \cdot \Delta S = \Delta H + W_{\text{useful}}$$

$$\text{or } -W_{\text{useful}} = \Delta H - T \cdot \Delta S \quad (7)$$

Also, from Gibb's free energy,

$$\Delta G = \Delta H - T \cdot \Delta S \quad (8)$$

From eqn (7) and (8)

$$-W_{\text{useful}} = (\Delta G)_{T,P}$$

$$(-\Delta G)_{T,P} = W_{\text{useful}} \quad (9)$$

Hence, decrease in Gibb's free energy is amount of useful work done at constant Temp. and Pressure.



V.V  
Imp

# Effect of temperature and spontaneity of process  
OR

# Effect of temp<sup>r</sup> on endothermic and exothermic  
OR

# Spontaneous process in the light of free energy change, enthalpy change and entropy change.

Spontaneous process can be defined with the help of free energy change, enthalpy change and entropy change. The equation is

$$\Delta G = \Delta H - T \cdot \Delta S$$

The value of  $\Delta G$  at different temperatures can be related by using these equation.

a) For exothermic process ( $\Delta H = -ve$ )

i) If  $\Delta S = +ve$ , then  $\Delta G = -ve$ , process is spontaneous at all temp<sup>r</sup>.

ii) If  $\Delta S = -ve$ , then  $\Delta G = +ve$

• At low temp<sup>r</sup>,  $\Delta H > T \cdot \Delta S$ , then  $\Delta G = -ve$

So process is spontaneous at low temp<sup>r</sup>.

• At high temp<sup>r</sup>,  $\Delta H < T \cdot \Delta S$ , then  $\Delta G = +ve$ , So process is non-spontaneous at high temp<sup>r</sup>.

b) For endothermic process ( $\Delta H = +ve$ )

i) If  $\Delta S = +ve$ , then  $\Delta G = -ve$ , process is non-spontaneous at all temp<sup>r</sup>.



i) At low temp<sup>r</sup>,  $\Delta H > T \cdot \Delta S$  then  $\Delta G = +ve$ , process is non-spontaneous at low temp<sup>r</sup>.

At high temp<sup>r</sup>,  $\Delta H < T \cdot \Delta S$  then  $\Delta G = -ve$ , process is spontaneous at high temp<sup>r</sup>.

ii) If  $\Delta S = -ve$ ,  $\Delta G = +ve$ , process is non-spontaneous at all temp<sup>r</sup>.

c) At equilibrium,  $\Delta H = T \cdot \Delta S$ , process is at equilibrium.

Relation between Gibb's free energy and equilibrium constant:

$$\Delta G = -RT \ln K$$

$$= -2.303 RT \log K$$

where,

$K$  = equilibrium constant

$R$  = universal gas constant

$T$  = Kelvin temp<sup>r</sup>.

Relation between Gibb's free energy and EMF of cell

$$\Delta G = -nFE_{\text{cell}}$$

where

$n$  = number of moles

$F$  = Faraday constant

$E_{\text{cell}}$  = Emf of cell

At Standard condition ( $25^\circ\text{C}$ , 1 atm)

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ$$